Exercises

Exercise 4.1 · Quadratics. Solve in $\mathbb{C}$: $z^2+9=0$; $z^2-4z+13=0$; $2z^2+2z+1=0$; $z^4-1=0$.

Solution. $z^2+9=(z-3i)(z+3i)$: $\{-3i, 3i\}$. Next $\Delta=-36$: $2-3i$ and $2+3i$. Then $\Delta=-4$: $\frac{-2\pm 2i}{4}$, that is $-\frac{1}{2}-\frac{1}{2}i$ and $-\frac{1}{2}+\frac{1}{2}i$. Finally $z^4-1=\left(z^2-1\right)\left(z^2+1\right)$: $\{1,-1,i,-i\}$.

Exercise 4.2 · Reducing to a quadratic.

1. Solve in $\mathbb{C}$ the equation $z^4+4z^2-5=0$ by setting $Z=z^2$.
2. Solve in $\mathbb{C}$ the equation $z+\dfrac{1}{z}=1$, for $z\neq 0$.

Solution.

1. $Z^2+4Z-5=(Z-1)(Z+5)$. So $z^2=1$ or $z^2=-5$: $\{1,-1,i\sqrt{5},-i\sqrt{5}\}$.
2. For $z\neq 0$, the equation is equivalent to $z^2-z+1=0$, with discriminant -3: $z=\frac{1-i\sqrt{3}}{2}$ or $z=\frac{1+i\sqrt{3}}{2}$, both nonzero.

Exercise 4.3 · An obvious root. Let $P(z)=z^3-3z^2+4z-2$.

1. Check that 1 is a root of P.
2. Determine the polynomial Q such that $P(z)=(z-1)Q(z)$.
3. Solve $P(z)=0$ in $\mathbb{C}$.

Solution.

1. $1-3+4-2=0$.
2. $(z-1)\left(z^2+bz+c\right)=z^3+(b-1)z^2+(c-b)z-c$, hence $b=-2$ and $c=2$: $Q(z)=z^2-2z+2$.
3. Q has discriminant -4: roots $1-i$ and $1+i$. The solutions are 1, $1-i$, $1+i$.

Exercise 4.4 · A non-real root. Let $P(z)=z^3+z^2+4z+4$.

1. Check that $2i$ is a root of P. Deduce another root without computation.
2. Expand $(z-2i)(z+2i)$, then factorise $P(z)$ and solve $P(z)=0$.

Solution.

1. $P(2i)=-8i-4+8i+4=0$. The coefficients are real, so $-2i=\overline{2i}$ is also a root.
2. $(z-2i)(z+2i)=z^2+4$ and $P(z)=z^2(z+1)+4(z+1)=(z+1)\left(z^2+4\right)$. The solutions are -1, $2i$, $-2i$.

Exercise 4.5 · Factorising $z^3 - 27$.

1. Factorise $z^3 - 27$ by $z - 3$ using the course.
2. Solve in $\mathbb{C}$ the equation $z^3 = 27$ and plot the solutions in the complex plane. What do you notice?

Solution.

1. $z^3 - 3^3 = (z-3)\left(z^2 + 3z + 9\right)$.
2. The quadratic has discriminant $9 - 36 = -27 = \left(3i\sqrt{3}\right)^2$: $z = \frac{-3 \pm 3i\sqrt{3}}{2}$. The three solutions 3, $\frac{-3-3i\sqrt{3}}{2}$, $\frac{-3+3i\sqrt{3}}{2}$ have modulus 3 and form an equilateral triangle inscribed in the circle with centre O and radius 3 (chapter 5).

Exercise 4.6 · A parameter. Let m be a real number. Consider the equation (E_m): $z^2 - 2z + m = 0$.

1. Depending on the value of m, give the number and the nature of the solutions of (E_m).
2. Assume $m > 1$. Let A and B be the images of the solutions. Determine m so that the triangle OAB is equilateral.

Solution.

1. $\Delta = 4 - 4m$. If $m < 1$: two real solutions $1 \pm \sqrt{1-m}$; if $m = 1$: one solution, 1; if $m > 1$: two conjugate solutions $1 \pm i\sqrt{m-1}$.
2. $OA = OB = \sqrt{1 + (m-1)} = \sqrt{m}$ and $AB = 2\sqrt{m-1}$. The triangle is equilateral if and only if $m = 4(m-1)$, that is $m = \frac{4}{3}$ (which satisfies $m > 1$).

Exercise 4.7 · Recovering a polynomial. Determine the polynomial P of degree 3, with real coefficients and leading coefficient 1, such that $1 + i$ is a root of P and $P(0) = 4$.

Solution. Since the coefficients are real, $1 - i$ is also a root. Factorise P by $z - (1+i)$; the quotient, of degree 2, vanishes at $1-i$ (because $1-i \neq 1+i$), so it can in turn be factorised by $z-(1-i)$. What remains is a factor of degree 1 with leading coefficient 1: $P(z) = (z-1-i)(z-1+i)(z-r) = \left(z^2 - 2z + 2\right)(z-r)$ for some complex number r. $P(0) = -2r = 4$, so $r = -2$ and $P(z) = \left(z^2 - 2z + 2\right)(z+2) = z^3 - 2z + 4$. Check: $P(0) = 4$ and $P(1+i) = (-2+2i) - 2 - 2i + 4 = 0$.

Problem

Problem · Coefficients and roots of a cubic polynomial. Let $P(z) = z^3 + az^2 + bz + c$ be a polynomial with complex coefficients, with three roots z_1, z_2, z_3 (not necessarily distinct), so that $P(z) = (z - z_1)(z - z_2)(z - z_3)$ for every complex number z.

1. By expanding, prove that
$$z_1 + z_2 + z_3 = -a, \qquad z_1z_2 + z_1z_3 + z_2z_3 = b, \qquad z_1z_2z_3 = -c.$$
2. Check these relations on the polynomial of the activity, $z^3 - 2z^2 + 4z - 8$.
3. Prove that $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. Consider $R(z) = z^3-2z^2+3z-5$, which we take for granted can be written $(z-z_1)(z-z_2)(z-z_3)$. Compute $z_1^2 + z_2^2 + z_3^2$ without solving the equation.
5. Deduce that the roots of R are not all real.

Beyond the syllabus · The d'Alembert-Gauss theorem. Every polynomial of degree $n \geq 1$ with complex coefficients factorises in $\mathbb{C}$ as a product of n factors of degree 1. This theorem is not part of the syllabus: it is admitted in question 4 and is not assessed.

Solution · Problem.

1. $(z - z_1)(z - z_2)(z - z_3) = z^3 - (z_1 + z_2 + z_3)z^2 + (z_1z_2 + z_1z_3 + z_2z_3)z - z_1z_2z_3$. Identifying the coefficients (property taken for granted) gives the three relations.
2. The roots are 2, $2i$, $-2i$: sum $2 = -(-2)$; sum of the pairwise products $4i - 4i + 4 = 4$; product $8 = -(-8)$.
3. $(z_1 + z_2 + z_3)^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_1z_3 + z_2z_3)$, so $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. $a = -2$ and $b = 3$: $z_1^2 + z_2^2 + z_3^2 = 4 - 6 = -2$.
5. If the three roots were real, the sum of their squares would be nonnegative. It equals -2: at least one root is not real.

Teacher note · Status of the problem. Vieta's formulas are a "possible problem" of the syllabus. The fact that every polynomial of degree 3 factorises in this way over $\mathbb{C}$ (the d'Alembert-Gauss theorem) is beyond the syllabus: it is taken for granted in the statement. For R, the associated real function is strictly increasing (its derivative $3x^2-4x+3$ has a negative discriminant): one real root, two non-real conjugate roots.